DIGITAL DETECTIVES

Spotting the Unreal in an AI World

Grade 6 | Session 1: Ethics of AI

CASE FILE #001: The Impossible Speech

Imagine watching a video of your favorite Indian actor. In just one minute, they speak fluently in **Hindi, Tamil, Telugu, Bengali, and Marathi!** Their lips move perfectly with every word, and their voice sounds exactly like it should.

"Wait a minute... does this actor really speak ten different languages fluently? Is this a real achievement, or are you looking at a digital mask?"

1. Understanding Deepfakes & AI Technology

What are Deepfakes? Deepfakes are videos or audio recordings created by Artificial Intelligence (AI) that look or sound real but are actually fake. Think of it as a *Digital Mask*. AI uses a computer to "wear" someone else's face or voice, hiding the real person underneath. Just like a magician uses tricks to make things disappear, AI uses math to make people say things they never actually said.

How Does AI Do It? AI creates deepfakes by analyzing thousands of existing videos and audio recordings. It learns exactly how a person's face moves, how their eyes blink, and how their voice sounds. This process involves three main steps:

- **Lip Movement Analysis:** Studying lip positions for every spoken syllable.
- **Voice Pattern Recording:** Capturing tone, pitch, pace, and unique vocal habits.
- **Expression Mapping:** Recreating subtle micro-expressions (smiles, eyebrow raises, blinks).

2. Detective Toolkit: Spotting the Glitches

✓ Real Media

- Natural eyes and regular blinking patterns.
- Smooth, consistent physical movements.
- Background remains crisp, sharp, and clear.
- Audio matches lip synchronization flawlessly.

X AI Deepfake

- Unnatural/stiff blinking or motionless eyes.
- Robotic, metallic, or unnatural vocal cadence.
- Odd flickering, blurring, or double shadows near face/hair.
- Glitches where the digital mask meets the background.

3. The Speed of Misinformation & Digital Responsibility

Misinformation refers to fake news or false messages spread online—often through messaging platforms like WhatsApp, social media feeds, or video channels. It travels fast: *"Like a digital wildfire, once you press 'forward', it's hard to take it back!"*

Real-World Example: A viral WhatsApp message claiming *"School is closed tomorrow due to heavy rain"* when official authorities have declared no such holiday.

Why Must We Be Careful?

1. **It's Hard to Take Back:** Information spreads exponentially. You cannot "un-send" a fake story once thousands have viewed it.
2. **It Can Cause Harm:** False rumors trigger unnecessary anxiety, confusion, and can hurt innocent people.
3. **Protect Your Circle:** Being a Digital Detective helps protect your friends and family from being tricked by *nakli* (fake) media.

THE DETECTIVE PLEDGE

"I will look closely for digital masks and check with adults before sharing.
I will not forward strange messages without proof and will help others spot fake news!"

DIGITAL DETECTIVES: STUDENT WORKSHEET & PRACTICE QUESTIONS

Session 1: Ethics of AI & Deepfake Verification

Part A: Detective Field Scenario Challenge

Scenario: Imagine you are scrolling online and see a viral video of a famous pop singer promoting a brand-new gaming app, claiming that downloading it gives free gift cards. The video looks convincing, but something feels slightly strange.

1. List three visual or auditory clues you should check in the video to determine if it is a real video or a deepfake.

2. Describe the exact verification steps you should take before sharing this video with your classmates or family.

Part B: Student Practice Questions (No Answers Included)

Q1. Conceptual Definition

In your own words, define what a "deepfake" is and explain why it is compared to a "Digital Mask."

Q2. Technical Process & AI Learning

How does an Artificial Intelligence model learn to replicate a specific person's voice and facial expressions? Mention two key elements involved in this learning process.

Q3. Real-World Misinformation Impact

Why is forwarding unverified news or messages compared to a "digital wildfire"? Give one concrete example of harm caused by forwarding fake school closure notifications.

Q4. Ethical Decision Making

If a classmate sends you a video that appears to show a teacher saying something funny but mean, what steps should you follow according to the "Digital Detective Pledge"?

Q5. Comparative Analysis Table

Identify whether each characteristic below belongs to **Real Media (R)** or **AI Deepfake (D)**:

Characteristic / Observation	Classification (R / D)
a) Natural eye blinking and smooth cheek motion	
b) Metallic audio tone with lip movements slightly delayed	
c) Blurry, flickering pixels around the jawline and background edges	
d) Clear lighting, natural background focus, and matched audio timing	